

Unit 5 :: Introduction to Probability

☞ PHY0400404: Mathematical Physics ☞

Dr. Parag Bhattacharya
Department of Physics
Rangapara College (Autonomous)

Syllabus

Unit 5: Introduction to Probability

Independent random variables, probability distribution functions; binomial, Gaussian and Poisson distributions with examples; mean and variance.

1 Fundamental concepts of probability

1.1 Random experiment

Any process of observation is referred to as an experiment. The results of an observation are called the outcomes of the experiment. *An experiment is called a random experiment if its outcome cannot be predicted.* Typical examples of a random experiment are as follows:

- i. the roll of a die
- ii. the toss of a coin
- iii. drawing a card from a deck
- iv. selecting a message signal for transmission from several messages

1.2 Sample space

The set of all possible outcomes of a random experiment is called the *sample space* (or *universal set*), and it is denoted by S .

An element in a sample space S is called a *sample point*. Each outcome of a random experiment corresponds to a sample point.

Examples

1. Find the sample space for the experiment of tossing a coin (a) once and (b) twice.

Solution:

- (a) There are two possible outcomes, heads (H) or tails (T). Thus:

$$S = \{H, T\}$$

- (b) There are four possible outcomes, which are pairs of heads and tails. Thus:

$$S = \{HH, HT, TH, TT\}$$

2. Find the sample space for the experiment of tossing a coin repeatedly and of counting the number of tosses required until the first head appears.

Solution: All possible outcomes for this experiment are the terms of the sequence $1, 2, 3, \dots$

Thus

$$S = \{1, 2, 3, \dots\}$$

Therefore, there are an infinite number of outcomes.

1.3 Event

An *event* is a subset A of the sample space S , i.e., it is a set of possible outcomes.

If the outcome of an experiment is an element of A , we say that the event A has occurred.

An event consisting of a single point of S is often called a *simple* or *elementary event*.

If A and B are events of a sample space S , then

1. $A \cup B$ is the event “either A or B or both” and is called the *union of A and B* .
2. $A \cap B$ is the event “both A and B ” and is called the *intersection of A and B* .
3. A' is the event “not A ” and is called the *complement of A* .
4. $A - B = A \cap B'$ is the event “ A but not B .” In particular, $A' = S - A$.

1.4 Probability

If an event can occur in h different ways out of a total number of n possible ways, all of which are equally likely, then the probability of the event is $\frac{h}{n}$.

1.4.1 The axioms of probability

Suppose we have a sample space S . If S is discrete, all subsets correspond to events and conversely, but if S is non-discrete, only special subsets (called measurable) correspond to events.

To each event A in the class C of events, we associate a real number $\Pr(A)$, which is called the *probability of the event A* , if the following axioms are satisfied.

- **Axiom 1:** For every event A in the class C ,

$$\Pr(A) \geq 0$$

- **Axiom 2:** For the sure or certain event S in the class C ,

$$\Pr(S) = 1$$

- **Axiom 3:** For any number of mutually exclusive events $A_1, A_2, \dots$, in the class C ,

$$\Pr(A_1 \cup A_2 \cup \dots) = \Pr(A_1) + \Pr(A_2) + \dots$$

2 Random variables

3 Probability distribution functions

3.1 Binomial distribution

3.2 Gaussian distribution

3.3 Poisson distribution

4 Statistics of a distribution

4.1 Mean of a distribution

4.2 Variance of a distribution